

(Gas Absorption)

Pre-Lab Plan

Unit Operations Lab 2

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Group 2, Tuesday, AM section

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1. Experimental Objective (5 pts)

The objective of this experiment is to study gas absorption in a packed-bed column by measuring packed-bed hydrodynamics and CO₂ mass transfer. Air, CO₂, and water flow rates are set, and pressure drop across the packing is measured using a red-oil manometer to identify the flooding point and an appropriate operating range.

The experiment begins by determining the packed-bed void fraction, then measuring pressure drop as a function of air and water flow rates to characterize flooding behavior. After operating conditions are selected, CO₂ absorption by water is studied in both a closed-loop and open-loop water system, and CO₂ desorption is observed after shutting off the CO₂ feed. The results are used to relate pressure-drop behavior and flow regime to overall absorber performance and mass-transfer trends.

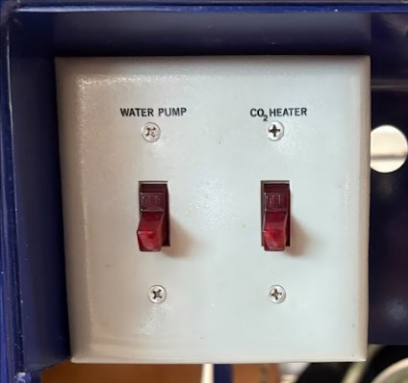
2. Experimental

2.1 Apparatus (5 pts)

In this experiment, there are vital pieces of equipment. These involve the column by which air, water, and CO₂ will enter, the water pump, multiple rotameters, multiple valves, and the water reservoir. The full list of equipment can be seen below:

Table 1 – Laboratory Equipment

Equipment	Number	Picture	Description
Water Pump	1		The water pump is used to bring water from the reservoir tank to the top of the column.

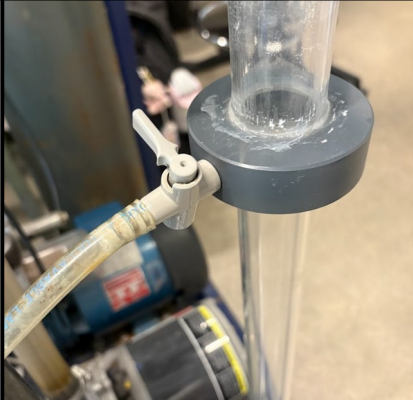
Electrical Switches	2		Switches used to shut on/off the water pump and CO₂ heater in the system.
Water Pump Discharge Valve	3		Valve used to direct water from the system into the drain.

Packed Column	4		A column packed with cylindrical pellets from which air, water, and CO₂ will travel through.
Air Supply Needle Valve & Gauge	5		A valve and gauge used to control and measure the pressure of air entering the system from utilities.

CO₂ Valve & Gauge	6		Valve and Gauge used to control and measure the CO₂ content entering the column.
Air Rotameter & Valve	7		Valve and measurement device used to control and measure the volumetric flowrate of air flowing through the column.

Water Rotameter & Valve	8	 The image shows a vertical water rotameter. At the top is a red valve handle with a white label 'C1'. Below the handle is a black pipe section. The main body of the rotameter is a clear, vertical tube with a scale from 0 to 10 L/min. The scale is marked with numbers 1 through 10. The fluid level is currently at approximately 8.5 L/min. The brand name 'Platon' is visible on the side of the tube. The bottom of the tube has a black base with a small white label '1.5'.	Valve and measurement device used to control and measure the volumetric flowrate of water flowing through the column.
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CO₂ Rotameter & Needle Valve	9		Valve and measurement device used to control and measure the volumetric flowrate of CO₂ flowing through the column.
Red Oil Manometer(s)	10		Manometers that measure the pressure difference from the bottom and top of the column in centimeters of red oil.

Manometer Air Pressure Bottom Valve	11		A valve that can be opened to take air pressure measurements from the bottom of the column to read pressure differentials in the manometer
Liquid Sample Valve	12		Valve that can be turned open to collect a sample of liquid from the column for analysis
Manometer Air Pressure Top Valve	13		A valve that can be opened to take air pressure measurements from the top of the column to read pressure differentials in the manometer.

CO₂ Measurement Device Flask	14		A flask that will be filled with sodium hydroxide to measure the amount of CO₂ transferred into the air from the column
Water Reservoir	15		A reservoir used to store water flowing through the system.
Recirculation Valve	16		A valve that controls the amount of water recirculating from the column back into the water reservoir

2.2 M

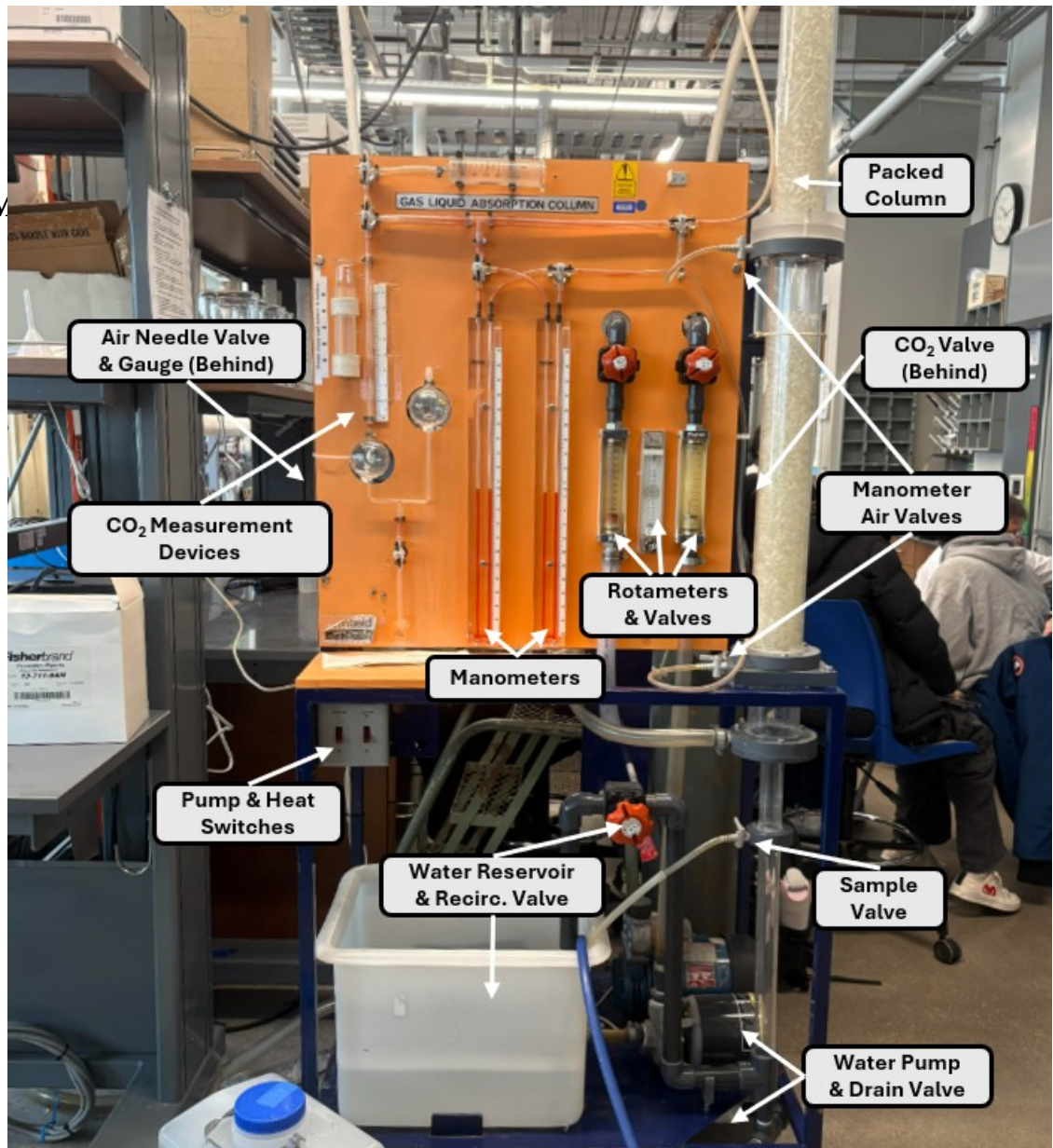
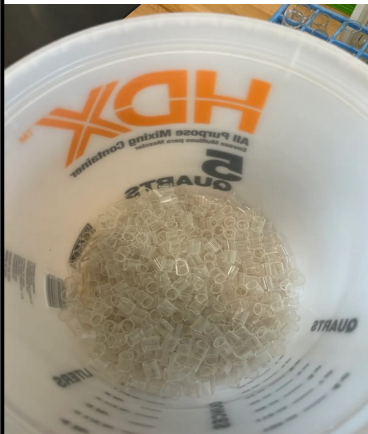
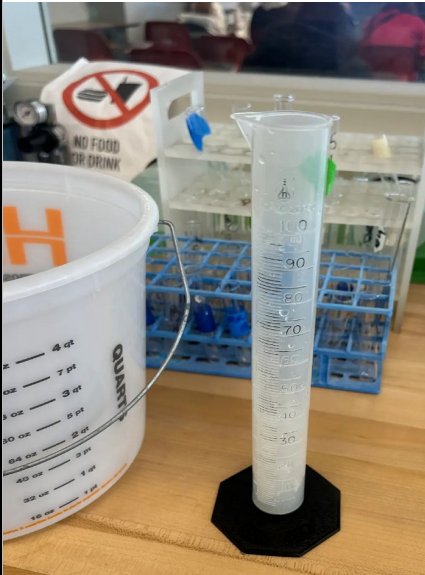
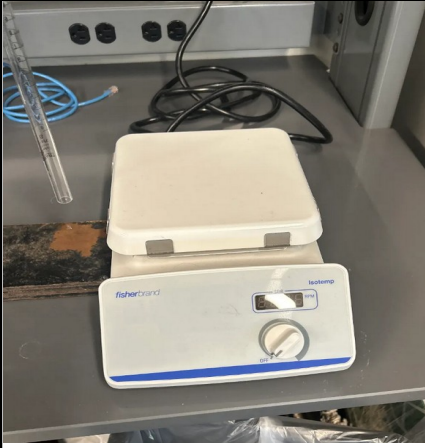


Figure 1 - Full Apparatus

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The complete material and supply list required for the Gas Absorption are provided in Table 2. This table includes all materials and supplies used in the experiment, along with brief details for each item.

Equipment	Number	Picture	Description
Digital Thermometer	1		Measures the temperature of deionized water in the reservoir during the experiment.

Deionized Water Supply Line	2		Used to fill the liquid reservoir tank with deionized water which serves as the absorbing liquid in the packed bed column.
Deionized Water Container	3		Containers are used to store and transport deionized water.
Plastic Packings	4		Packing material used to increase gas liquid contact and improve mass transfer
Safety Goggles	5		Protective equipment used to protect the eyes during the experiment

Graduated Cylinder	6		Used to measure liquid volumes and collect water samples for titration
Carbon Dioxide Gas Cylinder	7		Supplies CO₂ gas to the absorption column during the experiment.
Magnetic Hot Plate	8		Used during sample preparation or titration to mix solutions uniformly while heating if required.

Titration Stand	9		Stand used to hold burettes and other glassware during titration
Parafilm Sealing Film	10		Used to seal containers or flasks to prevent contamination or evaporation of samples.
Phenolphthalein Indicator	11		Indicator used to determine the titration endpoint for CO₂ analysis

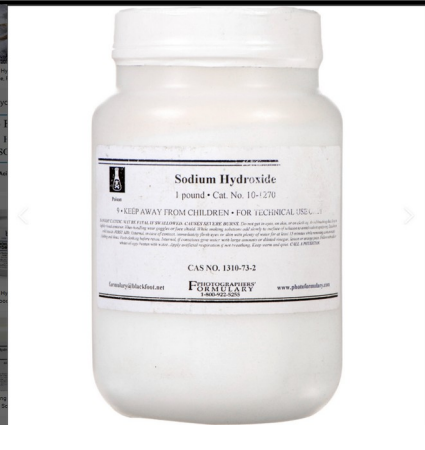
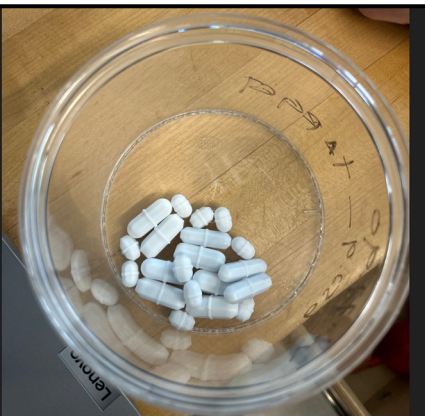
Sodium Hydroxide	12		Base solution used to titrate water samples for CO₂ concentration.
Compressed Air Supply Line	13		Provides the air stream for the packed bed absorption experiment.
Magnetic Stir Bars			Used with a magnetic stirrer to mix solutions during titration or sample preparation.

Table 2: Material and Supply List

2.3 Experimental Plan (10 pts)

Procedure

Part 1: Packed Bed Characterization

A. Void Volume Determination

1. Measure a known total volume of 1-cm Raschig ring packing using a graduated cylinder.
2. Slowly add deionized water to the packing until the void spaces are filled.
3. Record the volume of water required to fill the voids.
4. Calculate the bed porosity (void fraction) using:

$$\varepsilon = \frac{V_{void}}{V_{total}}$$

where V_{void} is the volume of water added and V_{total} is the total packed bed volume.

B. Pressure Drop and Flooding Determination

1. Close the drain valve under the liquid reservoir and close column drain valve D1.
2. Fill the liquid reservoir tank approximately three-quarters full with deionized water. Record the water temperature.
3. Open compressed air shut-off valve C2 and adjust airflow to **20 L/min** using control valve C5 (ensure valve C3 is closed).
4. Allow air to flow for 5 minutes to dry the packing.
5. Measure pressure drop across the packed bed using the red oil manometer for different air flowrates (no liquid flow).
6. Close valve C5 and start the liquid pump.
7. Open valve C4 and adjust water flow to **1.5 L/min** using flowmeter F1 and valve C1.
8. Adjust liquid level so it remains above sampling valve S4 but below the gas inlet tube.
9. Reopen gas valve C5 and increase airflow gradually from 20 L/min.
10. Record pressure drop as airflow increases until flooding is observed (sharp rise in pressure drop).

11. Repeat flooding measurements for water flowrates of:

- 3 L/min
- 6 L/min
- 8 L/min

Carefully monitor the manometer near flooding to prevent red oil from being pushed into tubing.

Part 2: Closed-Loop CO_2 Absorption and Desorption

Part 2A: CO_2 Absorption (Closed Loop System)

1. Close all drain valves and fill the reservoir three-quarters full with deionized water. Record temperature.
2. Collect an initial water sample (50 mL) and determine initial CO_2 concentration by titration.
3. Start the liquid pump and adjust water flow to **6 L/min**.
4. Open compressed air supply and adjust airflow to **20 L/min**. Record air pressure.
5. Open CO_2 cylinder valve fully and adjust regulator to ~20 psi.
6. Adjust CO_2 flow (2–5 L/min as assigned). Turn on heater block.
7. After 5 minutes, begin collecting 50 mL water samples from S4 at regular intervals.
8. Continue sampling until dissolved CO_2 concentration becomes constant.
9. Measure outlet gas CO_2 volume fraction using the Hempl apparatus at sample port S2.

Part 2B: CO_2 Desorption

1. Shut off CO_2 flow (close valve C3).
2. Reduce liquid flow to **1.5 L/min**.
3. Open drain valve D1 and close valve C4.
4. Maintain liquid level above S4 but below air inlet tube.
5. Immediately begin collecting water samples at ~1-minute intervals.
6. Continue titration analysis until CO_2 concentration decreases and stabilizes.

Part 3: Open-Loop CO_2 Absorption

1. Close drains and refill reservoir with deionized water. Record temperature.
2. Measure initial CO_2 concentration in water via titration.
3. Start pump and adjust water flow to **1.5 L/min**.
4. Open drain valve D1 and close C4 to operate in open-loop mode.
5. Adjust liquid level to remain above S4 but below air inlet.
6. Set airflow to **20 L/min**.
7. Adjust CO_2 flow (2–6 L/min as assigned). Turn on heater block.
8. Immediately begin sampling 50 mL water at:
 - o Every 30 seconds (first 2 minutes)
 - o Every 1 minute (next 5 minutes)
 - o Every 5 minutes until steady state
9. Once steady state is reached, measure outlet gas CO_2 concentration using the Hempl apparatus.

Variables

Independent Variables

- Air flowrate
- Water flowrate
- CO_2 flowrate
- Operating mode (closed vs open loop)

Controlled Variables

- Packing type (1-cm Raschig rings)
- Column geometry
- Working fluid (deionized water)
- Ambient temperature and pressure

Dependent Variables

- Pressure drop across packed bed
- Flooding point
- CO_2 concentration in liquid
- CO_2 volume fraction in outlet gas

- Overall mass transfer coefficient

Project Safety Evaluation (10 pts)

	<i>Yes/No</i>	
Potential electrical hazards?	Yes	If yes, identify conditions: Water is present near powered equipment, specifically the pump. Spills or leaks can create shock hazards and may damage electrical components.
Potential mechanical hazards?	Yes	If yes, identify conditions: Running the pump dry/unprimed can damage the pump internals.
Potential pressure hazards?	Yes	If yes, identify conditions: Carbon dioxide and compressed air are supplied at high pressure (e.g., ~750 psig CO ₂ cylinder and ~120 psig compressed air), then reduced to safer operating pressures using regulators (e.g., ~20 psig CO ₂ and ~40 psig air). Use regulators correctly, secure cylinders, and do not exceed the

		specified operating pressures.
Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	Yes	If yes, hazards involved: Carbon Dioxide (CO₂): Simple asphyxiant; can displace oxygen in poorly ventilated areas. Sodium hydroxide (NaOH): Strongly caustic/corrosive; causes severe skin/eye burns; harmful if mist/aerosol is inhaled. Phenolphthalein indicator: Irritant and flammable (alcohol-based indicator); avoid ignition sources and skin/eye contact. Red oil (manometer fluid): Flammable; if gas flow is too high it can be pushed into the manometer lines (dyeing/contaminating lines) and can leak, creating both housekeeping and fire concerns.
Gloves required?	Yes	If yes, specify type: Nitrile Gloves
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Deionized water , compressed air , carbon dioxide , red oil , sodium hydroxide , phenolphthalein
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	~30-40°C	CO ₂ in the supply tank is stored at high pressure and can be very cold. The gas first passes through a heater, which raises its temperature before it reaches the pressure regulator. After heating and pressure reduction, the CO ₂ stream is assumed to be approximately 30–40 °C (not directly measured). While warm, this temperature does not present a thermal hazard under normal operating conditions.
Pressure (psia)	~750 psia	CO ₂ is stored in a high-pressure cylinder (~750 psia) and is reduced to approximately 20 psig during operation using a pressure regulator. Compressed air is supplied at ~120 psig and reduced to ~40 psig before entering the system. Regulators must be verified before operation to prevent over-pressurization.
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
Water Control	yes	Fill and drain the system carefully and clean spills immediately to prevent slipping. If overfilling or leakage occurs, use the water shut-off valve, turn off the pump, or reduce air flow to slow or stop water flow before cleanup.
Electrical Control	yes	Keep electrical cords and powered components away from wet areas. If liquid contacts electrical equipment, shut off power at the main power switch before cleaning and restarting the system.
Chemical Control	yes	Wear nitrile gloves and eye protection when handling sodium hydroxide, phenolphthalein indicator, or red oil. Keep flammable indicator solutions and red oil away from ignition sources and clean spills promptly. Ensure the pump is properly primed before operation and do not

